

Startling secrets: Startle eye blink modulation by concealed crime information[☆]

Bruno Verschuere^{a,*}, Geert Crombez^a, Ernst H.W. Koster^a,
Bram Van Bockstaele^a, Armand De Clercq^b

^a *Department of Psychology, Ghent University, Henri Dunantlaan 2, B-9000 Ghent, Belgium*

^b *Department of Applied Mathematics and Computer Science, Ghent University, Krijgslaan 281 S9, B-9000 Ghent, Belgium*

Received 15 November 2006; accepted 17 June 2007

Available online 22 June 2007

Abstract

We used the startle eye blink paradigm to investigate the processes underlying physiological responding to concealed information. Autonomic responding was measured together with eye blink responses to startle probes presented during mock crime and control pictures. Based upon ‘orienting theory’, greater startle modulation to crime pictures in comparison to control pictures was expected. The electrodermal, heart rate and respiratory changes in Experiment 1 ($n = 29$) showed enhanced orienting to the crime pictures, but we observed reduced rather than enhanced startle modulation. In Experiment 2 ($n = 91$) and Experiment 3 ($n = 38$), participants either were or were not instructed to inhibit physiological responding in order to test whether inhibition was an explanation for the pattern of startle blink responding. Reduced startle modulation was observed only when participants inhibited physiological responding, confirming the proposed role of inhibition. The data suggest that not only orienting, but also inhibition contributes to physiological responding to concealed information.

© 2007 Elsevier B.V. All rights reserved.

Keywords: Startle eye blink; Attention; Concealed Information Test; Deception; Inhibition

Polygraph or “lie detector” tests are among the most commonly used psychological tests for the detection of deception. Polygraph tests aim to measure the enhanced cognitive and/or emotional reactions elicited by crime-related questions in comparison to control questions. There exist two main polygraph test formats: the Comparison Question Test format (also referred to as Control Question Technique) and the Concealed Information Test format (also referred to as Guilty Knowledge Test). These formats differ in the type of control questions used. In the Comparison Question Test format, physiological responses on crime-related questions (e.g., “Did you take the money from the bank?”) are compared to arousal-evoking comparison questions (e.g., “Did you ever take anything that did not belong to you?”). The test assumptions hold that the guilty suspect will show stronger reactivity to the

crime-related questions, whereas the innocent suspect is expected to show the largest response to the arousal-evoking comparison questions (Honts, 2005). In the Concealed Information Test, physiological responses to crime-related questions (e.g., “Was the victim shot?”) are compared to several incorrect control questions (e.g., “Was the victim...poisoned?...hanged?...drowned?...beaten to death?...strangled?”). The test assumptions hold that only the guilty suspect will show stronger orienting responses to the crime-related question compared to the incorrect control questions (Lykken, 1974).

The National Research Council (2003) reviewed the scientific evidence on polygraph testing, and concluded that “polygraph research has failed to build and refine its theoretical base (...). As a consequence, the field has not accumulated knowledge over time or strengthened its scientific underpinnings in any significant manner (p. 102)”. This is true for the Comparison Question Technique: there is virtually no scientific literature on the test theory or any empirical tests to falsify it. The Concealed Information Test, however, has been argued to have a solid theoretical underpinning (Ben-Shakhar and Elaad,

[☆] Parts of these data were presented at the 45th Annual Meeting of the Society for Psychophysiological Research (Lissabon, Portugal).

* Corresponding author. Tel.: +32 92648622; fax: +32 92646489.

E-mail address: Bruno.Verschuere@ugent.be (B. Verschuere).

2003). The orienting reflex may help to explain enhanced responding to concealed information (Lykken, 1974). The orienting reflex is the reaction to novel stimuli (e.g., a sudden noise), and is associated with behavioral and physiological changes that allow the organism to explore the novel stimulus. Orienting to this novel stimulus declines through repeated presentations; that is, habituation occurs. Familiar stimuli that are of known relevance to the organism (e.g., the door bell ringing) will also elicit an orienting reflex. Lykken (1974) argued that the crime-related questions in the Concealed Information Test are of known relevance only for the guilty suspect. Therefore, only the guilty suspect will show enhanced orienting to the crime-related questions. Empirical evidence has supported orienting theory in explaining the effects of the Concealed Information Test (for a review see Verschuere et al., 2004). Ben-Shakhar and Gati (2003), for example, have shown that skin conductance orienting in the Concealed Information Test is subject to habituation. The test theory was also refined through empirical work. Increasing the number of control alternatives relative to the number of crime-related questions leads to more differential responding (Lieblisch et al., 1970). Thus, not only relevance but also (relative) novelty contributes to orienting to the crime-related questions. Furthermore, a meta-analysis on 80 laboratory studies identified two other moderating factors: motivational instructions and overt deception (Ben-Shakhar and Elaad, 2003). It remains to be investigated whether these factors contribute to physiological responding by increasing the magnitude of the orienting reflex or by other processes independent of the orienting reflex (Ben-Shakhar and Elaad, 2003; Elaad and Ben-Shakhar, 1989).

The present study aimed to strengthen the theoretical base of the Concealed Information Test by using the startle eye blink. The startle eye blink is a component of the startle reflex that is elicited by intense stimuli, such as a burst of white noise at 105 dB(A). The size of this reflex is reliably modulated by a preceding lead stimulus (e.g., a sound or a picture) that itself does not elicit startle responding. The direction of this change is determined by the time interval between the onset of the lead stimulus and the onset of the startle probe. At short intervals (e.g., <500 ms), the startle reflex is inhibited. This “prepulse inhibition” is thought to reflect a protection of processing mechanism. At long intervals (e.g., >1000 ms), the startle reflex is facilitated, and this “prepulse facilitation” is thought to reflect general activation or attention (Filion et al., 1998). Of particular importance for the present study is the finding that orienting to the lead stimulus enhances startle modulation. Using a tone discrimination task, Filion et al. (1994) showed that orienting to the lead stimulus enhances prepulse inhibition at short intervals, and enhances prepulse facilitation at long lead intervals. In that study, participants were requested to count the longer than usual lead tones of one pitch (task relevant), and to ignore lead tones of a different pitch (task irrelevant). Startle eliciting bursts of noise were presented during tones of both pitches either 120 or 2000 ms after tone onset. The task relevant tones elicited stronger orienting compared to the task irrelevant tones, as indexed by an increase in skin conductance. It was found that the task relevant tones elicited stronger inhibition at

120 ms, and stronger facilitation at 2000 ms, compared with the to be ignored task irrelevant tones (for a review see Filion et al., 1998). Enhanced startle modulation was also found when visual rather than auditory lead stimuli were used. Lipp et al. (1998), for example, modified the discrimination task by Filion et al. (1994). These authors found enhanced prepulse facilitation to task relevant pictures compared to task irrelevant pictures at 3500–4500 ms. Furthermore, the enhanced startle modulation to orienting stimuli is not restricted to the discrimination task. Stronger inhibition at 240 ms and stronger facilitation at 3500 ms during interesting Gothic compared to boring Roman letters in participants who passively attended the letter strings (Lipp et al., 2001). These data indicate that orienting to the lead stimulus enhances startle modulation (i.e., greater startle inhibition at short lead intervals and greater startle facilitation at long lead intervals).

The orienting theory of the Concealed Information Test predicts enhanced startle modulation to crime pictures in comparison to the control pictures. Using a within-subjects design, participants acquired knowledge of one crime, and were ignorant of other crimes. Details from the crime that they had executed, served as crime-relevant items, and details of the crime that they were innocent of, served as control items. Specifically, we presented participants with pictures of both crimes for 6 s. Two-third of the pictures were probed with an acoustic startle probe, either after 300 ms or after 3000 ms. Skin conductance, heart rate, and respiration line length were also recorded.

1. Experiment 1

1.1. Method

1.1.1. Participants

Twenty-nine undergraduate students (19 female) of Ghent University took part as partial fulfilment of course requirements. Due to experimenter and/or equipment failure, data were lost for startle eye blink ($n = 6$), skin conductance ($n = 1$), and respiration ($n = 2$).

1.1.2. Experimental stimuli

The experimental stimuli in the polygraph test were 12 digital color pictures (height = 95 mm; width = 127 mm), consisting of 6 crime details of each crime (e.g., a bank note of €10). The blink-eliciting stimulus was a 50 ms 105 dB(A) burst of white noise with near immediate (<1 ms) rise time, generated by a V85-05C Coulbourn audio module, and provided binaurally by a Phillips headphone.

1.1.3. Response scoring and analysis

Psychophysiological signals were recorded by a Coulbourn Labline V, gated through a Scientific Solutions DMA card to a PC, and digitized using a customized program called Psychophysiological Recording (PSPHR). The psychophysiological data were later analyzed off line using Psychophysiological Analysis (PSPHA; De Clercq et al., 2006).

The eye blink component of the startle reflex was measured following guidelines provided by Blumenthal et al. (2005). Two miniature Ag/AgCl (.5 cm diameter) electrodes, filled with high conductive gel, were positioned over the orbicularis oculi muscle of the left eye. Skin was first cleaned with an alcohol preparation, and then a thin layer of gel was massaged into the recording site. One electrode was placed just below the left pupil, and the other placed 1 cm laterally, with the ground placed on the forehead. The raw electromyographic signal was amplified 50,000 times, filtered (band pass: 13–1000 Hz; and using a built-in 60 Hz notch filter), and digitally stored at 1000 Hz. Using PSPHA, we calculated the magnitude of the eye blink EMG by subtracting the mean rectified baseline value (0–20 ms after probe onset) from the rectified peak

value in the 21–200 ms interval after probe onset. PSPHA gave warnings of possible artefacts whenever the baseline value deviated more than three standard deviations from the participant's mean baseline level. Based on visual inspection, trials were rejected when the baseline was noisy, or a blink occurred just before probe onset. Startle responses during the pictures are expressed as a percentage change score compared to the startle responses during the inter-trial interval, using the formula: $((EMG_{\text{PICTURE}} - EMG_{\text{ITI}})/EMG_{\text{ITI}}) \times 100$ (Filion et al., 1994). A composite score for early and late startle eye blink modification was calculated (i.e., percentage change at 3000 ms – percentage change at 300 ms). A score of 0 would indicate no startle modulation. The amount of facilitation at 3000 ms and the amount of inhibition at 300 ms both increase the score: the higher the score, the greater the startle modulation.

Skin conductance was measured using a constant voltage (.5 V) coupler, and Ag/AgCl electrodes (.8 cm diameter) filled with KY-jelly that were attached on the thenar and hypothenar eminences of the left hand. Skin conductance was digitized at 10 Hz. Using PSPHA, we calculated skin conductance response as a baseline to peak difference score. Baseline was measured at one second before picture onset, and peak was measured as maximal skin conductance amplitude in the .5–5 s after picture onset (with a minimum of .05 μ S). Negative values were scored as zero. Data were square root transformed prior to analysis (Dawson et al., 2000).

An ECG was recorded by using three electrodes placed in lead II configuration. The ECG was filtered (band pass: 8–40 Hz) and digitized at 500 Hz. PSPHA was used to detect the *r*-peaks in the ECG and to calculate the distance between them. An artifact detection and correction procedure was applied with PSPHA. Prior to analysis, the interbeat intervals (IBI) were converted to heart rate in beats per minute (bpm) per real-time epoch (1 s). Mean bpm in the 3 s preceding stimulus onset were compared to the mean bpm in the 6 s period after stimulus onset.

Respiration was measured using a single strain gauge attached around the thorax. Respiration moved the air in the elastic tube and these variations were picked up by a pressure sensor. The difference in pressure was converted to voltage and digitized at 250 Hz. In line with applied research on the physiological detection of deception, we calculated respiration line length (see e.g., Ben-Shakhar and Dolev, 1996). The length of the respiration line was measured starting from stimulus onset up to 10 s later and expressed in arbitrary units. Timm (1982) has pointed out that the length of the respiration line might be disproportionally affected by the start of measurement. For example, starting at the end of a slowly declining expiratory curve or at the beginning of the rapidly ascending inspiratory curve would produce different line lengths for the same time interval. In order to deal with this problem, each respiration line length was calculated as the mean of 10 respiration line lengths: from stimulus onset to 10 s later, from .1 s after stimulus onset to 10.1 s after stimulus onset, from .2 to 10.2 s after stimulus onset, etc. We calculated respiration line length with PSPHA using the following formula: respiration line length = $\text{SQRT}(dX^2 + dY_1^2) + \text{SQRT}(dX^2 + dY_2^2) + \dots + \text{SQRT}(dX^2 + dY_n^2)$ with

- dX being the difference in time (expressed in milliseconds) between two subsequent data points.
- dY being the difference in amplitude (expressed in AD-values) between two subsequent data points.
- n being the number of data points (after smoothing) in the time period under investigation, i.e., 10 s.

1.1.4. Procedure

The mock crime procedure was identical to Verschuere et al. (2004). In short, participants were told that they would take part in a lie detection experiment, and they were instructed to feign innocence in a subsequent polygraph test. Participants then undertook either a mock theft or a mock fraud. The 15 participants enacting the mock theft were instructed to steal €10 from a coat. The 14 participants enacting the mock fraud were asked to copy the answers from an examination paper. Both mock crimes took place in rooms that students know they are forbidden to access. After execution of the mock crime, participants returned to the laboratory and physiological devices were attached. Participants were seated approximately 50 cm from the computer screen. Participants were first presented with the pictures ($n = 12$; 6 of each crime) that were to be presented in the Concealed Information Test. These pictures were shown for 5 s in a random order, with an inter-stimulus interval (ISI; here

defined as the time between the offset of a picture and the onset of the following picture) of 1000 ms. Participants were asked to look at all the pictures. This presentation is standard practice in field applications of the Concealed Information Test (see e.g., Hira and Furumitsu, 2002), and does not affect detection efficiency (Verschuere and Crombez, submitted). Next, participants received three presentations of the startle probe, because large habituation is typically found during the first presentations.

Next, the polygraph test followed. Participants were told that their primary task was to beat the polygraph by trying to conceal recognition of crime pictures. Motivational instructions on self-esteem were given: it was told that despite its high accuracy, intelligent people are able to beat the polygraph. Simultaneously with picture onset, the question “Do you know whether this picture is related to the crime?” appeared just above the picture. Participants were requested to give an overt “no” response at picture offset. The Concealed Information Test started with a buffer item (picture of a pen), followed by 36 pictures (18 crime, and 18 control), presented in three blocks of 12 pictures each. Each block consisted of six pictures of each crime, presented in the middle of the screen during 6 s, with an ISI from 18 to 24 s. The auditory startle probe was presented on two thirds of the pictures, 300 or 3000 ms after picture onset. The remaining third of the pictures were presented without probe in order to avoid probe anticipation/habituation. Pictures were presented in one of four fixed semi-random orders, with the restriction that there could be no more than three consecutive probed (versus unprobed) pictures, and no more than three consecutive pictures of one crime. In addition to the probes presented during the pictures, in each block four startle probes were presented in the middle of the ISI between two pictures.

Immediately after the Concealed Information Test, memory for the pictures was assessed. Pictures were presented at random one by one in the middle of the screen. Participants indicated whether they recognized the pictures as crime relevant or not by pressing one of two buttons on the keyboard.

1.2. Results

Hypotheses were tested using paired sample *t*-tests. We calculated Hedges and Olkin (1985) estimator of unbiased effect size *d*. Following the criteria provided by Cohen (1988), a *d* of .20, .50 and .80 was used as thresholds to define small, medium and large effects, respectively.

1.2.1. Memory check

Participants classified 96% of the pictures correctly.

1.2.2. Startle blink, skin conductance, respiration line length, and heart rate change

Startle modulation was reduced on crime pictures compared to the control pictures, $t(22) = 2.97$, $p < .01$, $d = .67$ (see Table 1). Skin conductance responses were greater to crime pictures than to control pictures, $t(27) = 2.56$, $p < .05$, $d = .23$ (see Table 2). Respiration line length was shorter after crime than after control pictures, $t(26) = 3.32$, $p < .01$, $d = .17$. Heart rate was lower after crime than after control pictures, $t(28) = 2.71$, $p < .05$, $d = .36$.

1.3. Discussion

Enhanced orienting to the crime pictures compared to the control pictures was evident in skin conductance, heart rate and respiration. Startle modulation for crime pictures was reduced instead of enhanced in comparison to the control pictures. The reduced startle modulation was unexpected and is difficult to reconcile with the orienting hypothesis. Note that the reduced startle modulation is unlikely to be due to the emotional valence

Table 1

Mean percentage startle modulation (standard deviations in parentheses) to crime and control pictures relative to startle responses during the ITI under the inhibition and view instruction in Experiments 1–3

	300 ms lead interval		3000 ms lead interval		Composite score	
	Crime	Control	Crime	Control	Crime	Control
Experiment 1	–17.94 (28.98)	–30.84 (18.44)	.34 (47.12)	18.44 (49.72)	18.28 (41.39)	49.28 (49.21)
Experiment 2						
View	–10.29 (27.44)	–12.29 (24.65)	6.84 (18.40)	5.93 (24.50)	17.14 (30.48)	18.22 (27.09)
Inhibition	–13.30 (28.83)	–17.68 (30.85)	–.59 (21.20)	11.05 (37.05)	12.71 (30.58)	28.73 (40.57)
Experiment 3						
View	–5.58 (25.95)	–5.38 (18.40)	7.80 (25.98)	.85 (32.54)	13.38 (29.73)	6.23 (35.50)
Inhibition	–12.74 (31.51)	–24.96 (25.98)	–7.39 (25.67)	–4.30 (25.08)	5.34 (34.25)	20.65 (24.11)

Note: The composite score was calculated by subtracting the percentage change at 300 ms from the percentage change at 3000 ms. The more startle modulation, the higher the composite score.

Table 2

Mean (standard deviations in parentheses) autonomic responses to crime and control pictures in Experiments 1–3

	SCR (μ S)		HR change (bpm)		RLL (arbitrary units)	
	Crime	Control	Crime	Control	Crime	Control
Experiment 1	.28 (.24)	.23 (.19)	.01 (1.69)	.65 (1.80)	10,628 (249)	10,673 (273)
Experiment 2						
View	.23 (.25)	.19 (.19)	–.70 (1.50)	–.03 (1.50)	10,761 (299)	10,791 (312)
Inhibition	.23 (.24)	.22 (.19)	–1.36 (1.39)	–.08 (1.46)	10,620 (348)	10,675 (357)
Experiment 3						
View	.18 (.18)	.16 (.17)	–1.06 (1.77)	–.58 (1.76)	10,287 (265)	10,294 (277)
Inhibition	.20 (.17)	.19 (.19)	–3.10 (2.13)	–1.06 (1.77)	10,270 (260)	10,272 (272)

Note: SCR = skin conductance response (in microsiemens or μ S); HR change = heart rate change (in beats per minute or bpm); RLL = respiration line length (in arbitrary units).

of the pictures (Bradley et al., 1993). Provided that crime pictures were perceived as more threatening than control pictures, enhanced startle modulation would be expected (particularly at the long lead interval). Thus, the reduced startle modulation cannot be explained easily in terms of attentional or emotional startle modulation. Given the evidence in support of the orienting hypothesis and the results of the other autonomic measures (i.e., respiratory and heart rate deceleration), one might argue that some task characteristics could explain the reversed effects. One explanation is that reduced startle modulation was observed because the (visual) lead stimuli and the (auditory) startle stimuli differed in sensory modality. It has been argued that attention effects may be modality specific (Anthony and Graham, 1985). However, this hypothesis was empirically falsified (Lipp et al., 2003). Thorne et al. (2005) concluded that the cross modality hypothesis does not hold in tasks with discrete stimuli. It seems unlikely that the reduced startle modulation in our study is due to the cross modal presentation of the stimuli. Experiment 2 served to (1) find additional evidence for the orienting hypothesis, and (2) test alternative explanations for the reduced startle modulation.

2. Experiment 2

The skin conductance, respiratory and heart rate responding to crime pictures observed in Experiment 1 supports the

orienting theory. To obtain further support for the orienting hypothesis, we assessed memory for the pictures immediately after the Concealed Information Test. Orienting predicts more elaborate processing of the crime pictures, hence better memory consolidation in comparison to the control pictures (Iacono et al., 1984). We could not simply assess memory of the pictures, because participants have prior experience with the crime but not the control pictures. To address this problem, we added some new objects to the pictures in the Concealed Information Test. Memory for these previously unseen details on the crime and control pictures was assessed, allowing an unbiased test of memory. This memory test also allows ruling out the possibility that participants diverted attention away from (i.e., avoided) the crime pictures during the Concealed Information Test. Avoidance of the crime pictures would result in impaired recognition of the details on the crime pictures compared to the control pictures.

Although orienting theory may explain most findings in the Concealed Information Test, other processes may be involved in the detection of concealed information. The reduced startle modulation for the crime pictures in Experiment 1 requires a different explanation. Concealing information involves inhibition, which may contribute to responding to concealed information. Of particular relevance to the present study is the finding by Jackson et al. (2000) that inhibition leads to reduced startle blinks at long lead intervals (see also, Dillon and

LaBar, 2005). We hypothesized that inhibition may account for the reduced startle modulation for the crime pictures. To test this hypothesis, we included a group of participants who were not required to inhibit responding to the crime pictures. All participants enacted a mock crime and were engaged in a Concealed Information Test. Half of the participants were asked to inhibit responding to the crime pictures, and the remaining half were asked to passively attend to all pictures. If inhibition explains the reduced startle modulation, reduced startle modulation should emerge only in participants who inhibit physiological responding.

2.1. Method

2.1.1. Participants

Ninety-one undergraduate students (56 female) of Ghent University participated and were rewarded either with course credits or €9. Due to experimenter and/or equipment failure, data were lost for startle eye blink (inhibition instruction: $n = 2$; view instruction: $n = 4$).

2.1.2. Procedure

The same mock crime procedure from Experiment 1 was used, but a number of methodological changes were made. First, a between subjects factor was created. Half of the participants received traditional polygraph instructions, requesting them to inhibit responding to the crime pictures (inhibition instruction: $n = 44$). The other participants were told that they formed a control condition and were asked to merely look at the pictures (view instruction: $n = 47$). To avoid inhibition in the view instruction condition, the sentence “Do you know whether this picture is related to the crime?” above each picture was omitted and participants were no longer requested to give an overt “no” answer. Second, memory for the crime and control pictures was assessed after the Concealed Information Test. To allow unbiased recall we added four irrelevant objects to all pictures in the Concealed Information Test. For example, the picture of the stolen €10 now also included a pen, some paperclips, a can of soda and an envelope. A free recall test, in which participants were asked to name as many details of the pictures as possible, was first administered. The subsequent cued recall test consisted of one multiple-choice question for each picture with eight possible objects. Participants were asked to indicate only those objects that they felt confident they had seen in the specific picture. For both the free and the cued recall task, the dependent variable was the total number of correctly named objects for each picture type (MIN = 0, MAX = 24).

2.2. Results

2.2.1. Memory

The 2 (instruction: inhibition versus view) $\times$ 2 (picture: crime versus control) mixed ANOVA on the free recall test showed that participants remembered more of the details of the crime pictures ($M = 6.01$, $S.D. = 2.57$) in comparison to the control pictures ($M = 4.78$, $S.D. = 2.78$), $F(1, 88) = 18.93$, $p < .001$, $d = .48$. This effect was greater with the inhibition instruction (crime: $M = 6.07$, $S.D. = 2.50$ versus control: $M = 4.27$, $S.D. = 2.55$) than for the view instruction (crime: $M = 5.96$, $S.D. = 2.66$ versus control: $M = 5.17$, $S.D. = 2.95$), although this was not significant $F(1, 88) = 2.92$, $p = .09$. Results for the cued recall test also showed better recall for the details of the crime pictures ($M = 9.63$, $S.D. = 3.27$) in comparison to the control pictures ($M = 8.15$, $S.D. = 3.02$), $F(1, 89) = 18.65$, $p < .001$, $d = .47$, with no effects of instruction, $F's < 1$.

2.2.2. Startle blink, skin conductance, respiration line length, and heart rate change

Results were analyzed using a multivariate analysis of variance (MANOVA) with repeated measures treated as variates (Vasey and Thayer, 1987). Significant interaction effects were followed by paired sample t -tests.

The 2 (instruction: inhibition versus view) $\times$ 2 (picture: crime versus control) MANOVA on startle modulation revealed marginally significant effects of picture type, $F(1, 83) = 3.84$, $p = .05$, $d = .26$, and of instruction $\times$ picture type, $F(1, 83) = 2.93$, $p = .09$, $d = .37$ (see Table 1). Startle modulation to crime pictures was significantly smaller than to control pictures in the inhibition instruction condition, $t(42) = 2.43$, $p < .05$, $d = .44$, but not in the view instruction condition, $t(41) < 1$, $d = .04$.

Skin conductance responses were marginally greater to crime pictures compared to the control pictures, $F(1, 89) = 2.83$, $p = .09$, $d = .11$ (see Table 2). Respiration line length was shorter after crime than after control pictures, $F(1, 89) = 14.68$, $p < .001$, $d = .12$. Heart rate was lower after crime than after control pictures, $F(1, 89) = 25.23$, $p < .001$, $d = .65$. Except for a tendency for greater overall respiratory suppression under the inhibition instruction, $F(1, 89) = 3.55$, $p = .06$, $d = .39$, there were no effects with instruction as a factor, $F's < 1$.

2.3. Discussion

Experiment 2 strengthened the findings of Experiment 1. Compared to the control pictures, crime pictures elicited stronger respiratory suppression and heart rate deceleration. There was a tendency towards greater skin conductance responding to crime pictures than to control pictures. In further support of the orienting theory, both the free and the cued recall test showed better recall for crime pictures compared to control pictures. The inhibition instruction had no effect on these measures. In contrast, there was an effect of inhibition instruction on startle eye blink responding. When participants passively viewed the pictures in the Concealed Information Test, no effect of picture type on startle eye blink was obtained. When asked to inhibit responding, reduced startle modulation for crime pictures was observed, replicating the effects of Experiment 1. The use of the view instruction condition also allows exclusion of an alternative formulation of attentional startle modulation. One could argue that both picture types in our study were task relevant. Lipp et al. (2003) found no differential startle modulation between interesting and boring letters when participants had to count the number of vowels. These authors argued that the requirement to count the vowels made all letters task relevant, thereby overriding the effects of interesting versus boring font type. Such an effect would have led to enhanced startle modulation under the view instructions, and no startle modulation under the inhibition instruction in our study. Instead, we found no startle modulation under the view instruction, and reduced startle modulation under the inhibition instruction. These findings indicate that inhibition modulated startle eye blink in our experiments.

Experiments 1 and 2 revealed a dissociation between response measures. Orienting seems to account for differential responding in memory, skin conductance, respiration and heart rate, whereas inhibition seems to account for differential startle responding. However, it remains intriguing why a process would affect one response measure and not another. Whereas the effects of startle eye blink point to the role of inhibition, no stronger differential responding in recall, skin conductance, heart rate and respiration under the inhibition instruction was found. We decided to readdress this research question using a more powerful within-subjects design.

3. Experiment 3

Experiment 3 investigated the effects of orienting and inhibition on differential responding to crime and control pictures on several response measures. Similar to Jackson et al. (2000), we manipulated the inhibition versus view instruction within-subjects. Participants received instructions to either view or inhibit responding to the crime pictures on a trial by trial basis.

3.1. Method

3.1.1. Participants

Thirty-eight undergraduate students (29 female) of Ghent University participated and were rewarded either with course credits or €9. Due to experimenter and/or equipment failure, data were lost for skin conductance ($n = 1$), heart rate ($n = 1$), and respiration ($n = 2$).

3.1.2. Procedure

The method of Experiment 3 is similar to Experiment 2. Participants were told that they were involved in a lie detection experiment, examining how individuals can cope with the polygraph. They were further informed that crime and control pictures would be preceded by the instruction to either simply attend to the picture or to inhibit their responses to the crime picture. Thus, crime and control pictures were preceded by the written instruction (presented for 5 s) to either simply “view” (“KIJK” in Dutch) the following picture or to “inhibit” (“ONDERDRUK” in Dutch) physiological responding elicited by the crime pictures. Because of this additional manipulation, we increased the number of trials from 36 to 60 (30 crime, 30 control), with the ISI between the offset of a picture and the onset of the instruction screen for the subsequent picture varying between 15 and 19 s. Half of the pictures were preceded by the inhibition instruction, and half by the view instruction. As in Experiments 1 and 2, two-third of the pictures contained a startle probe (half 300 ms after picture onset, and half 3000 ms after picture onset), and one-third of the pictures did not contain a probe. Because of the increase in the number of trials we made two further adjustments to keep the session within 1 h. First, participants looked at a videotaped mock crime (Iacono et al., 1984) instead of actually committing a mock crime. All participants watched a videotaped mock crime (duration = 4 min) that consisted of a theft of jewels. Participants were asked to pay close attention to the video and to try to imagine that they committed the mock theft. The five crime pictures were details from this videotaped mock crime (e.g., jewels). The five control pictures came from a similar mock crime video that the participant had not seen. There were four control videos (theft of digital camera, €100, MP3-player, cellular phone). The choice of picture was randomly determined for each participant. Second, the recall tests after the polygraph examination were omitted, and so no additional objects were presented on the pictures in the Concealed Information Test.

3.2. Results

3.2.1. Memory check

Due to programming error the memory data of the first 15 participants were lost. The remaining 24 participants classified 96% of the pictures correctly.

3.2.2. Startle blink, skin conductance, respiration line length, and heart rate change

The 2 (instruction: inhibition versus view) $\times$ 2 (picture: crime versus control) MANOVA on startle modulation revealed a significant instruction $\times$ picture type interaction, $F(1, 37) = 5.19$, $p < .05$, $d = .21$ (see Table 1). Other F 's < 1.3 . Startle modulation to crime pictures was smaller compared to control pictures in the inhibition instruction condition, $t(37) = 2.97$, $p < .01$, $d = .51$, but not in the view instruction condition, $t(37) < 1.1$, $d = -.22$.

For skin conductance, a significant main effect of instruction was found, with greater skin conductance responses after the inhibition instruction compared to the view instruction, $F(1, 36) = 4.81$, $p < .05$, $d = .17$ (see Table 2). Other F 's < 1 .

For respiration line length a significant main effect of instruction was found, with shorter respiration line lengths after the inhibition instruction compared to the view instruction, $F(1, 35) = 9.42$, $p < .01$, $d = .07$. Other F 's < 1 .

For heart rate, the significant main effects of picture type, $F(1, 36) = 12.01$, $p < .01$, $d = .51$, and instruction, $F(1, 36) = 35.29$, $p < .001$, $d = 1.01$, resorted under the marginally significant picture type $\times$ instruction effect, $F(1, 36) = 3.76$, $p = .06$, $d = .37$. This interaction effect indicates that the greater heart rate deceleration to crime compared to control pictures was more pronounced when participants were instructed to inhibit responding (crime: $M = -3.10$, S.D. = 2.13 versus control: $M = -1.94$, S.D. = 1.86) than when instructed to merely look at the pictures (crime: $M = -1.06$, S.D. = 1.77 versus control: $M = -.58$, S.D. = 1.76).

3.3. Discussion

The instruction to inhibit physiological responding led to reduced startle modulation for the crime pictures compared to the control pictures. No significant differences in blink modulation between crime and control pictures were found when participants merely looked at the pictures. This indicates that startle eye blink was solely determined by inhibition in our set of studies, replicating the results of Experiments 1 and 2. Regarding skin conductance, heart rate, and respiration, we expected to find differential responding to crime and control pictures due to enhanced orienting, and reasoned that the inhibition instruction could further increase this differential responding. This pattern was found for heart rate, but not for skin conductance and respiration where only inhibition had an effect on differential responding.

3.4. General discussion

To test whether orienting accounts for enhanced responding to concealed information, we measured startle eye blink

responses for crime and control pictures in a Concealed Information Test. Previously examined responses (memory, skin conductance, heart rate, respiration line length) were also recorded. Supporting orienting theory, respiration line length was shorter after crime pictures than after control pictures (Experiments 1 and 2), recall (Experiment 2) was better for crime than for control pictures, and heart rate was lower after crime pictures than after control pictures (Experiments 1–3). Two findings require further clarification. First, although heart rate was lower after crime pictures compared to control pictures, no absolute deceleration compared to the pre-stimulus baseline (i.e., heart rate change below 0) was observed for the crime pictures in Experiment 1. We think that the modest increase in heart rate results from giving an overt answer (Raskin and Hare, 1978). In support of this argument, we observed an absolute decrease in heart rate when participants were no longer requested to give an overt answer (Experiments 2 and 3). Second, an important limitation of this study is the failure to find reliable differences in skin conductance between crime and control pictures. Previous studies on the Concealed Information Test have found large effects of electrodermal responding (i.e., $d = 1.55$; Ben-Shakhar and Elaad, 2003). The difference in skin conductance responding between crime and control pictures was significant in Experiment 1, but only marginally significant in Experiment 2 and non-significant in Experiment 3. To allow an examination of the mechanisms of the Concealed Information Test, our paradigm deviated in several ways from the optimal Concealed Information Test. First, we used a 1/1 proportion of crime/control pictures instead of the typical 1/4 proportion to allow an unbiased examination of the processes contributing to responding in the Concealed Information Test. In that way, crime and control pictures only differ in relevance and not in relative novelty. A disadvantage of the 1/1 proportion, however, is that it diminishes differential responding between crime and control pictures (Elaad and Ben-Shakhar, 1989; Lieblisch et al., 1970; Verschuere et al., 2004). Second, the use of the startle eye blinks may have further diminished differential responding in skin conductance. The arousal value of the pictures may be relatively small compared to that of the startle probes. It could be pointed out that previous research with concurrent presentation of startle probes did find effects of attention and emotion on skin conductance. The difference between our study and those about attention is that lead stimuli in attention research have to be attended to for adequate task performance (for an exception see, Lipp et al., 2001). The difference between our study and those about emotion is that the arousal value of the affective pictures in emotion research (e.g., pointed gun, spider, barking dog) is high. Third, the use of a silent answer test in Experiments 2 and 3 may have further decreased differential responding (Ben-Shakhar and Elaad, 2003). Despite these notes of caution, the reliable differences in heart rate, respiration and recall between crime and control pictures confirm the validity of the Concealed Information Test in this study. Orienting theory also predicted greater startle modulation for crime compared to control pictures. However, we consistently found reduced startle modulation for crime pictures when participants were asked to

inhibit physiological responding, and no startle modulation when participants were asked to simply look at the pictures. As explained above, neither the difference in sensory modality between the lead and the startle stimuli, the task relevance of the crime pictures, the emotional valence of the crime pictures, nor avoidance away from the crime pictures can explain these findings. Inhibition may account for the reduced startle modulation for crime pictures. In direct support of this hypothesis, no startle modulation was found when participants simply looked at the pictures in the Concealed Information Test. These data suggest that startle eye blink responding to the crime pictures was affected by inhibition and not by orienting.

The startle results indicate that inhibition contributes to responding to concealed information, and raise the question to what extent inhibition can explain autonomic responding to concealed information. One possibility is that inhibition per se (and not orienting) explains responding to concealed information. Indeed, Pennebaker and Chew (1985) theorized that inhibition demands effort and comes with a physiological cost. However, defining inhibition as “the executive function that allows one to intentionally inhibit a dominant, automatic or prepotent response” (Miyake et al., 2000), there needs to be a response in order to have inhibition. The orienting reflex seems a likely candidate to be the “dominant, automatic or prepotent response” in the Concealed Information Test. In the domain of thought suppression and emotion suppression, it was examined whether inhibition per se or rather inhibition of arousing events leads to autonomic arousal. The physiological cost of inhibiting exciting versus neutral thoughts was examined by Wegner et al. (1990). Skin conductance level was measured while undergraduates were asked to think or not think about sex versus neutral events (i.e., the weather). The results showed that inhibition of thoughts on sex, but not neutral events, increased skin conductance level. Gross and Levenson (1997) examined autonomic responding to emotionally arousing (sad or amusing) versus neutral film clips under natural viewing and suppression conditions. Inhibition led to increased sympathetic activity of the cardiovascular system for the emotionally arousing, but not the neutral film clips. These data show that inhibition comes with a physiological cost, but only for arousing events. More recently it was shown that these arousal-evoking events do not need to be emotional in nature. Hagemann et al. (2006) demonstrated that attempts to inhibit physiological arousal that was produced by a startle probe increased sympathetic arousal. Applied to the Concealed Information Test, an attempt to inhibit the physiological arousal that accompanies the orienting reflex may increase the very arousal one wishes to reduce. However, it should be acknowledged that our data do not uniformly support this reasoning. In Experiment 2, there were no incremental effects of inhibition above orienting for respiration, heart rate and memory. In Experiment 3, orienting and inhibition seemed to contribute to heart rate responding. For skin conductance and respiration, however, only inhibition produced significant effects. The present data do not allow a conclusive answer on how orienting and inhibition interact during a Concealed Information Test. Our results do suggest that inhibition may play a more

prominent role in the Concealed Information Test than is generally assumed. Moreover, results from Experiment 3 question whether mere recognition is sufficient for differential responding in the Concealed Information Test.

The idea that inhibition contributes independently to physiological responding in the Concealed Information Test may seem to contradict previous work that has demonstrated differential responding under minimal conditions. For example, even when participants remain silent and no overt deception is required, concealed information can still be detected above chance (cf. Experiment 2; for a review see Ben-Shakhar and Elaad, 2003). Furthermore, innocent “aware” examinees – who have acquired crime knowledge (e.g., as a witness) – show enhanced physiological responding to the crime items (Bradley et al., 1996). From studies under such minimal conditions, it has been concluded that mere recognition is sufficient for enhanced responding to concealed information (Ben-Shakhar and Furedy, 1990). This conclusion, however, requires two important qualifications. First, it can be questioned whether enhanced physiological responding in these studies is due to mere recognition and not to inhibition. Examinees are told that they take part in lie detection research, and are asked to (and often motivated to) try to appear innocent in the polygraph examination. Thus, it is not unlikely that participants still try to inhibit responding, even under minimal conditions. Second, even if inhibition would be found not to contribute under minimal conditions this would not exclude its role in a more realistic test procedure. Moreover, studies looking at behavioral and brain activity in detection of deception confirm the role of inhibition in the Concealed Information Test. Pennebaker and Chew (1985) monitored eye movements and facial expression during a Concealed Information Test. Deceptive responses were associated with an increase in skin conductance level and a decrease in facial expressive behavior. Recent brain imaging work is also suggestive of the role of inhibitory processes. Several studies have found increased activation of the anterior cingulate cortex and the dorsolateral prefrontal cortex during deception. These regions of the brain have been associated with a variety of executive functions, amongst them response inhibition (Phan et al., 2005). An interesting new approach to examine the role of inhibition is the experimental manipulation of inhibitory processes, for example by using rapid transcranial magnetic stimulation (rTMS). Selective activation or reduction of inhibitory processes by rTMS could shed more light on the role of inhibition for differential responding in the Concealed Information Test (Karim et al., 2006).

The meta-analysis by Ben-Shakhar and Elaad (2003) showed that overt denial can enhance differential responding in the Concealed Information Test. The present study extends this work by (1) proposing a mechanism (i.e., inhibition) for the effect of deception, (2) demonstrating in a more rigorous way that inhibition contributes to the Concealed Information Test, and (3) pointing out the need for a re-examination of whether recognition alone is sufficient for responding in the Concealed Information Test.

Acknowledgement

The authors wish to thank Helena Campens and Sam De Laet for their aid in data collecting.

References

- Anthony, B.J., Graham, F.K., 1985. Blink reflex modification by selective attention—evidence for the modulation of automatic processing. *Biological Psychology* 21, 43–59.
- Ben-Shakhar, G., Dolev, K., 1996. Psychophysiological detection through the guilty knowledge technique: effects of mental countermeasures. *Journal of Applied Psychology* 81, 273–281.
- Ben-Shakhar, G., Elaad, E., 2003. The validity of psychophysiological detection of information with the Guilty Knowledge Test: a meta-analytic review. *Journal of Applied Psychology* 88, 131–151.
- Ben-Shakhar, G., Furedy, J.J., 1990. Theories and Applications in the Detection of Deception. Springer-Verlag, New York.
- Ben-Shakhar, G., Gati, I., 2003. The effects of serial position and frequency of presentation of common stimulus features on orienting response reinstatement. *Psychophysiology* 40, 139–145.
- Blumenthal, T.D., Cuthbert, B.N., Filion, D.L., Hackley, S., Lipp, O.V., Van Boxtel, A., 2005. Committee report: guidelines for human startle eyeblink electromyographic studies. *Psychophysiology* 42, 1–15.
- Bradley, M.M., Cuthbert, B.N., Lang, P.J., 1993. Pictures as prepulse—attention and emotion in startle modification. *Psychophysiology* 30, 541–545.
- Bradley, M.T., MacLaren, V.V., Carle, S.B., 1996. Deception and nondeception in guilty knowledge and guilty actions polygraph tests. *Journal of Applied Psychology* 81 (2), 153–160.
- Cohen, J., 1988. *Statistical Power Analysis for the Behavioural Sciences*. Lawrence Erlbaum, Hillsdale.
- Dawson, M.E., Schell, A.M., Filion, D.L., 2000. The electrodermal system. In: Cacioppo, L.G.T.J.T., Berntson, G.B. (Eds.), *Handbook of Psychophysiology*. Cambridge University Press, NY, pp. 200–223.
- De Clercq, A., Verschuere, B., De Vlieger, P., Crombez, G., 2006. Psychophysiological analysis (PSPA): a modular script based program for analyzing psychophysiological data. *Behavior Research Methods* 38 (3), 504–510.
- Dillon, D.G., LaBar, K.S., 2005. Startle modulation during conscious emotion regulation is arousal-dependent. *Behavioral Neuroscience* 119, 1118–1124.
- Elaad, E., Ben-Shakhar, G., 1989. Effects of motivation and verbal response type on psychophysiological detection of information. *Psychophysiology* 26 (4), 442–451.
- Filion, D.L., Dawson, M.E., Schell, A.M., 1994. Probing the orienting response with startle modification and secondary reaction-time. *Psychophysiology* 31, 68–78.
- Filion, D.L., Dawson, M.E., Schell, A.M., 1998. The psychological significance of human startle eyeblink modification: a review. *Biological Psychology* 47, 1–43.
- Gross, J.J., Levenson, R.W., 1997. Hiding feelings: the acute effects of inhibiting negative and positive emotion. *Journal of Abnormal Psychology* 106 (1), 95–103.
- Hagemann, T., Levenson, R.W., Gross, J.J., 2006. Expressive suppression during an acoustic startle. *Psychophysiology* 43, 104–112.
- Hedges, L.V., Olkin, I., 1985. *Statistical Methods for Meta-analysis*. Academic Press, San Diego, CA.
- Hira, S., Furumitsu, I., 2002. Polygraphic examinations in Japan: applications of the guilty knowledge test in forensic investigations. *International Journal of Police Science and Management* 4, 16–27.
- Honts, C.R., 2005. The psychophysiological detection of deception. In: Strömwall, P.A.G.L.A. (Ed.), *The Detection of Deception in Forensic Contexts*. University Press, Cambridge, pp. 103–123.
- Iacono, W.G., Boisvenu, G.A., Fleming, J.A., 1984. Effects of diazepam and methylphenidate on the electrodermal detection of guilty knowledge. *Journal of Applied Psychology* 69, 289–299.

- Jackson, D.C., Malmstadt, J.R., Larson, C.L., Davidson, R.J., 2000. Suppression and enhancement of emotional responses to unpleasant pictures. *Psychophysiology* 37, 515–522.
- Karim, A.A., Lotze, M., Schneider, M., Dockery, C., Weber, C., Braun, C., et al., 2006. The truth about lying: inhibition of the anterior prefrontal cortex improves deceptive behavior. *Psychophysiology* 43, S50–S150.
- Lieblisch, I., Kugelman, S., Ben-Shakhar, G., 1970. Efficiency of GSR detection of information as a function of stimulus set size. *Psychophysiology* 6, 601–608.
- Lipp, O.V., Blumenthal, T.D., Adam, A.R., 2001. Attentional modulation of blink startle at long, short, and very short lead intervals. *Biological Psychology* 58, 89–103.
- Lipp, O.V., Neumann, D.L., Pretorius, N.R., McHugh, M.J., 2003. Attentional blink modulation during sustained and after discrete lead stimuli presented in three sensory modalities. *Psychophysiology* 40, 285–290.
- Lipp, O.V., Siddle, D.A.T., Dall, P.J., 1998. Effects of stimulus modality and task condition on blink startle modification and on electrodermal responses. *Psychophysiology* 35, 452–461.
- Lykken, D.T., 1974. Psychology and the lie detector industry. *American Psychologist* 29, 725–739.
- Miyake, A., Friedman, N.P., Emerson, M.J., Witzki, A.H., Howerter, A., Wager, T.D., 2000. The unity and diversity of executive functions and their contributions to complex “frontal lobe” tasks: a latent variable analysis. *Cognitive Psychology* 41 (1), 49–100.
- National Research Council, 2003. The Polygraph and Lie Detection. Committee to Review the Scientific Evidence on the Polygraph. The National Academies Press, Washington.
- Pennebaker, J.W., Chew, C.H., 1985. Behavioral-inhibition and electrodermal activity during deception. *Journal of Personality and Social Psychology* 49, 1427–1433.
- Phan, K.L., Magalhaes, A., Ziemlewicz, T.J., Fitzgerald, D.A., Green, C., Smith, W., 2005. Neural correlates of telling lies: a functional magnetic resonance imaging study at 4 T. *Academic Radiology* 12 (2), 164–172.
- Raskin, D., Hare, R.D., 1978. Psychopathy and detection of deception in a prison sample. *Psychophysiology* 15, 126–136.
- Thorne, G.L., Dawson, M.E., Schell, A.M., 2005. Attention and prepulse inhibition: the effects of task-relevant, irrelevant, and no-task conditions. *International Journal of Psychophysiology* 56, 121–128.
- Timm, H.W., 1982. Analyzing deception from respiratory patterns. *Journal of Police Science and Administration* 10, 47–51.
- Vasey, M.W., Thayer, J.F., 1987. The continuing problem of false positives in repeated measures Anova in psychophysiology—a multivariate solution. *Psychophysiology* 24 (4), 479–486.
- Verschuere, B., Crombez, G., submitted. Deja Vu! The effect of previewing test items on the accuracy of the Concealed Information Test.
- Verschuere, B., Crombez, G., De Clercq, A., Koster, E.H.W., 2004. Autonomic and behavioral responding to concealed information: differentiating orienting and defensive responses. *Psychophysiology* 41, 461–466.
- Wegner, D.M., Shortt, J.W., Blake, A.W., Page, M.S., 1990. The suppression of exciting thoughts. *Journal of Personality and Social Psychology* 58 (3), 409–418.